

Sketches: You can find the skeleton of the website with initial visualizations under the website folder on this git as well in the EDA.ipynb file. Some further analyses on player evolution have been performed.

Tools and lectures:

We will rely on lectures 6.1 (Perception, Colour) and 6.2 (Mark and Channels) to determine the colour schemes we should use for our visualisations. We will use the good and bad practices that we should apply in our website's visualisations detailed in lectures 7.1 (Design for Data Viz) and 7.2 (Do's and Don'ts). Lectures 8.1 (Maps) and 8.2 (Maps in practice) will also be helpful in the visualisation of the shots position on the basketball court. Finally lecture 12.1 (Storytelling) should be leveraged to create a good backbone for the website, by telling a story that would spark interest in the website visitors.

Goals and ideas:

We want to implement at least the four following ideas, as they would represent a MVP.

1. A spatial representation of the shots taken in the court, through the years, to identify the changes in profile, and other tendencies.
2. Graphical representation of the shot profile of some of the players, along with some statistics about how many shots they took, and some success measurements.
3. Spatial representation of the points in the court where a shot is usually the most successful (based on the number of points, the probability of achieving a clean shot...).
4. Spatial representation of all the shots that were taken in the last 5 seconds of Q4, with an indicator of whether the shot was successful or not. For this visualisation we also want to allow the user to filter based on a specific team and/or player. We should also display some statistics like how many shots were made given the selected filters.
5. Understand if teams shoot differently and see how the winning teams take their shots.
6. Following point 5, how does a player shooting profile evolve when changing teams ?

Tools used for each visualization:

As presented in the course we used a lot of [D3.js](#) for nice 3D rendering. Here below we detail more precisely the tools used for each section of the skeleton website used so far.

1. **Court / Hero Visualization (4.4 Million Shots)**
 - **Three.js:** We picked this for smooth, high-performance canvas rendering. It handles the massive 4.4 million shots in the browser without slowing things down.
 - **Custom Code within Three.js:** This is custom logic built in to manage the complex movement of all those individual shots, like the staggered falling effects and making sure shots made versus missed have the right color.
2. **Revolution (The Death of the Mid-Range)**

- **D3.js:** This draws the hexagon cells for the hexbin map. It's key for managing the data as we switch years, making sure the cell colors instantly update to show if a location's efficiency is above or below the league average.
 - **Vanilla JS:** We're using standard JavaScript to control the timeline slider, which manages the frame updates and coordinates with D3.
3. **Fingerprints (Player Shot Profiles)**
 - **D3.js:** This handles the complex math to create the radial bar chart. It maps how often a player shoots to the length of the bar, and their Field Goal percentage to an accurate color on the scale.
 4. **Value (Where Should You Shoot?)**
 - **D3.js:** Used to create the heat map grid over the court using SVG rectangles. D3 makes sure the data changes smoothly when a user switches between views like "Expected Value," "FG%," and "Shot Volume."
 5. **Clutch (Final Seconds)**
 - **D3.js:** This renders the interactive scatter plot. It handles filtering all the clutch shots by player or team, and it manages the hover-state popups (tooltips) that appear when you interact with a specific shot.

General UI and Utilities:

- **js/court.js:** This file uses D3.js utility features to set up the necessary scales and draw the standard NBA court boundaries that appear behind nearly all the visualisations.
- **Standard HTML5/CSS3:** We rely on these for the overall page structure, managing scroll-based reveals, and controlling overlay elements (we intentionally avoided heavy frontend frameworks like React or Vue).

Extra Ideas:

Finally, some extra and more challenging ideas we would like to add to the website would be to have an even more dynamic shot visualization, where the user could not only see where the shot was taken from but also an animation of the shot going in from the position it was taken. That is, take the orange dots currently on the website and make them go through the basket one by one in a smooth manner.

Also, another idea is to model shot selection under score pressure. That is, to create a filter to compare shot distributions (volume and efficiency) when a team is winning versus when they are losing by a tight margin (e.g., 1-5 points) in the 4th quarter. This would require augmenting the dataset with game score data, but it would provide valuable behavioral insight into risk tolerance and strategic choices during crucial moments